9 An Introduction to Elliptic Functions

> The form that Jacobi had given to the theory of elliptic functions was far from perfection; its flaws are obvious. At the base we find three fundamental functions sn, cn and dn. These functions do not have the same periods...
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> In Weierstrass' system, instead of three fundamental functions, there is only one, $\wp(u)$, and it is the simplest of all having the same periods. It has only one double infinity; and finally its definition is so that it does not change when one replaces one system of periods by another equivalent system.
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> *H. Poincaré,* 1899

The theory of elliptic functions, which is of interest in several parts of mathematics, initially grew out of the study of elliptic integrals. These can be described generally as integrals of the form $\int R(x, \sqrt{P(x)})\,dx$, where R is a rational function and P a polynomial of degree three or four.[1] These integrals arose in computing the arc-length of an ellipse, or of a lemniscate, and in a variety of other problems. Their early study was centered on their special transformation properties and on the discovery of an inherent double-periodicity. We have seen an example of this latter phenomenon in the mapping function of the half-plane to a rectangle taken up in Section 4.5 of the previous chapter.

It was Jacobi who transformed the subject by initiating the systematic study of doubly-periodic functions (called elliptic functions). In this theory, the theta functions he introduced played a decisive role. Weierstrass after him developed another approach, which in its initial steps is simpler and more elegant. It is based on his $\wp$ function, and in this chapter we shall sketch the beginnings of that theory. We will go as far as to glimpse a possible connection with number theory, by considering the Eisenstein series and their expression involving divisor functions. A number of more direct links with combinatorics and number theory arise from the theta

[1]The case when P is a quadratic polynomial is essentially that of "circular functions", and can be reduced to the trigonometric functions $\sin x$, $\cos x$, etc.

functions, which we will take up in the next chapter. The remarkable facts we shall see there attest to the great interest of these functions in mathematics. As such they ought to soften the harsh opinion expressed above about the imperfection of Jacobi's theory.

1 Elliptic functions

We are interested in meromorphic functions f on $\mathbb{C}$ that have two periods; that is, there are two non-zero complex numbers ω_1 and ω_2 such that

$$f(z+\omega_1) = f(z) \quad \text{and} \quad f(z+\omega_2) = f(z),$$

for all $z \in \mathbb{C}$. A function with two periods is said to be **doubly periodic**.

The case when ω_1 and ω_2 are linearly dependent over $\mathbb{R}$, that is $\omega_2/\omega_1 \in \mathbb{R}$, is uninteresting. Indeed, Exercise 1 shows that in this case f is either periodic with a simple period (if the quotient ω_2/ω_1 is rational) or f is constant (if ω_2/ω_1 is irrational). Therefore, we make the following assumption: the periods ω_1 and ω_2 are linearly independent over $\mathbb{R}$.

We now describe a normalization that we shall use extensively in this chapter. Let $\tau = \omega_2/\omega_1$. Since τ and $1/\tau$ have imaginary parts of opposite signs, and since τ is not real, we may assume (after possibly interchanging the roles of ω_1 and ω_2) that $\mathrm{Im}(\tau) > 0$. Observe now that the function f has periods ω_1 and ω_2 if and only if the function $F(z) = f(\omega_1 z)$ has periods 1 and τ, and moreover, the function f is meromorphic if and only if F is meromorphic. Also the properties of f are immediately deducible from those of F. We may therefore assume, without loss of generality, that f is a meromorphic function on $\mathbb{C}$ with periods 1 and τ where $\mathrm{Im}(\tau) > 0$.

Successive applications of the periodicity conditions yield

$$f(z+n+m\tau) = f(z) \quad \text{for all integers } n, m \text{ and all } z \in \mathbb{C}, \tag{1}$$

and it is therefore natural to consider the lattice in $\mathbb{C}$ defined by

$$\Lambda = \{n + m\tau : \ n, m \in \mathbb{Z}\}.$$

We say that 1 and τ **generate** Λ (see Figure 1).

Equation (1) says that f is constant under translations by elements of Λ. Associated to the lattice Λ is the **fundamental parallelogram** defined by

$$P_0 = \{z \in \mathbb{C} : z = a + b\tau \ \text{ where } 0 \leq a < 1 \text{ and } 0 \leq b < 1\}.$$

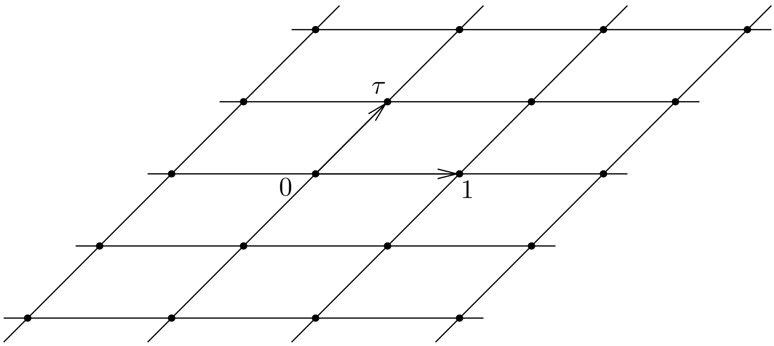

Figure 1. The lattice Λ generated by 1 and τ

The importance of the fundamental parallelogram comes from the fact that f is completely determined by its behavior on P_0. To see this, we need a definition: two complex numbers z and w are **congruent modulo** Λ if

$$z = w + n + m\tau \quad \text{for some } n, m \in \mathbb{Z},$$

and we write $z \sim w$. In other words, z and w differ by a point in the lattice, $z - w \in \Lambda$. By (1) we conclude that $f(z) = f(w)$ whenever $z \sim w$. If we can show that any point in $z \in \mathbb{C}$ is congruent to a unique point in P_0 then we will have proved that f is completely determined by its values in the fundamental parallelogram. Suppose $z = x + iy$ is given, and write $z = a + b\tau$ where $a, b \in \mathbb{R}$. This is possible since 1 and τ form a basis over the reals of the two-dimensional vector space $\mathbb{C}$. Then choose n and m to be the greatest integers $\leq a$ and $\leq b$, respectively. If we let $w = z - n - m\tau$, then by definition $z \sim w$, and moreover $w = (a - n) + (b - m)\tau$. By construction, it is clear that $w \in P_0$. To prove uniqueness, suppose that w and w' are two points in P_0 that are congruent. If we write $w = a + b\tau$ and $w' = a' + b'\tau$, then $w - w' = (a - a') + (b - b')\tau \in \Lambda$, and therefore both $a - a'$ and $b - b'$ are integers. But since $0 \leq a, a' < 1$, we have $-1 < a - a' < 1$, which then implies $a - a' = 0$. Similarly $b - b' = 0$, and we conclude that $w = w'$.

More generally, a **period parallelogram** P is any translate of the fundamental parallelogram, $P = P_0 + h$ with $h \in \mathbb{C}$ (see Figure 2).

Since we can apply the lemma to $z - h$, we conclude that every point in $\mathbb{C}$ is congruent to a unique point in a given period parallelogram. Therefore, f is uniquely determined by its behavior on any period parallelogram.

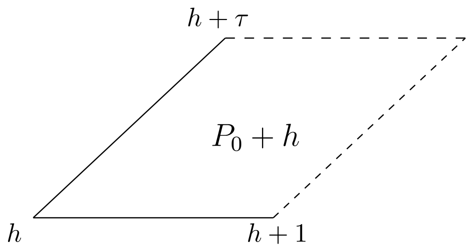

Figure 2. A period parallelogram

Finally, note that Λ and P_0 give rise to a covering (or tiling) of the complex plane

$$\mathbb{C} = \bigcup_{n,m\in\mathbb{Z}} (n + m\tau + P_0), \tag{2}$$

and moreover, this union is disjoint. This is immediate from the facts we just collected and the definition of P_0. We summarize what we have seen so far.

Proposition 1.1 *Suppose f is a meromorphic function with two periods 1 and τ which generate the lattice Λ. Then:*

(i) *Every point in $\mathbb{C}$ is congruent to a unique point in the fundamental parallelogram.*

(ii) *Every point in $\mathbb{C}$ is congruent to a unique point in any given period parallelogram.*

(iii) *The lattice Λ provides a disjoint covering of the complex plane, in the sense of* (2)*.*

(iv) *The function f is completely determined by its values in any period parallelogram.*

1.1 Liouville's theorems

We can now see why we assumed from the beginning that f is meromorphic rather than just holomorphic.

Theorem 1.2 *An entire doubly periodic function is constant.*

Proof. The function is completely determined by its values on P_0 and since the closure of P_0 is compact, we conclude that the function is bounded on $\mathbb{C}$, hence constant by Liouville's theorem in Chapter 2.

A non-constant doubly periodic meromorphic function is called an **elliptic function**. Since a meromorphic function can have only finitely many zeros and poles in any large disc, we see that an elliptic function will have only finitely many zeros and poles in any given period parallelogram, and in particular, this is true in the fundamental parallelogram. Of course, nothing excludes f from having a pole or zero on the boundary of P_0.

As usual, we count poles and zeros with multiplicities. Keeping this in mind we can prove the following theorem.

Theorem 1.3 *The total number of poles of an elliptic function in P_0 is always ≥ 2.*

In other words, f cannot have only one simple pole. It must have at least two poles, and this does not exclude the case of a single pole of multiplicity ≥ 2.

Proof. Suppose first that f has no poles on the boundary ∂P_0 of the fundamental parallelogram. By the residue theorem we have

$$\int_{\partial P_0} f(z)\,dz = 2\pi i \sum \operatorname{res} f,$$

and we contend that the integral is 0. To see this, we simply use the periodicity of f. Note that

$$\int_{\partial P_0} f(z)\,dz = \int_0^1 f(z)\,dz + \int_1^{1+\tau} f(z)\,dz + \int_{1+\tau}^{\tau} f(z)\,dz + \int_{\tau}^{0} f(z)\,dz,$$

and the integrals over opposite sides cancel out. For instance

$$\begin{aligned}
\int_0^1 f(z)\,dz + \int_{1+\tau}^{\tau} f(z)\,dz &= \int_0^1 f(z)\,dz + \int_1^0 f(s+\tau)\,ds \\
&= \int_0^1 f(z)\,dz + \int_1^0 f(s)\,ds \\
&= \int_0^1 f(z)\,dz - \int_0^1 f(z)\,dz \\
&= 0,
\end{aligned}$$

and similarly for the other pair of sides. Hence $\int_{\partial P_0} f = 0$ and $\sum \operatorname{res} f = 0$. Therefore f must have at least two poles in P_0.

If f has a pole on ∂P_0 choose a small $h \in \mathbb{C}$ so that if $P = h + P_0$, then f has no poles on ∂P. Arguing as before, we find that f must have at least two poles in P, and therefore the same conclusion holds for P_0.

The total number of poles (counted according to their multiplicities) of an elliptic function is called its **order**. The next theorem says that elliptic functions have as many zeros as they have poles, if the zeros are counted with their multiplicities.

Theorem 1.4 *Every elliptic function of order m has m zeros in P_0.*

Proof. Assuming first that f has no zeros or poles on the boundary of P_0, we know by the argument principle in Chapter 3 that

$$\int_{\partial P_0} \frac{f'(z)}{f(z)}\, dz = 2\pi i(\mathcal{N}_{\mathfrak{z}} - \mathcal{N}_{\mathfrak{p}})$$

where $\mathcal{N}_{\mathfrak{z}}$ and $\mathcal{N}_{\mathfrak{p}}$ denote the number of zeros and poles of f in P_0, respectively. By periodicity, we can argue as in the proof of the previous theorem to find that $\int_{\partial P_0} f'/f = 0$, and therefore $\mathcal{N}_{\mathfrak{z}} = \mathcal{N}_{\mathfrak{p}}$.

In the case when a pole or zero of f lies on ∂P_0 it suffices to apply the argument to a translate of P.

As a consequence, if f is elliptic then the equation $f(z) = c$ has as many solutions as the order of f for every $c \in \mathbb{C}$, simply because $f - c$ is elliptic and has as many poles as f.

Despite the rather simple nature of the theorems above, there remains the question of showing that elliptic functions exist. We now turn to a constructive solution of this problem.

1.2 The Weierstrass $\wp$ function

An elliptic function of order two

This section is devoted to the basic example of an elliptic function. As we have seen above, any elliptic function must have at least two poles; we shall in fact construct one whose only singularity will be a double pole at the points of the lattice generated by the periods.

Before looking at the case of doubly-periodic functions, let us first consider briefly functions with only a single period. If one wished to construct a function with period 1 and poles at all the integers, a simple choice would be the sum

$$F(z) = \sum_{n=-\infty}^{\infty} \frac{1}{z+n}.$$

Note that the sum remains unchanged if we replace z by $z+1$, and the poles are at the integers. However, the series defining F is not absolutely